PolicyGraphAudit-RT

Compliance Audit Report
Heterogeneous GNN Privacy Analysis

Notably Text Editor - Writer &

`com.notably.texteditor.note`

Developer	DEHA
Genre	TOOLS
Installs	10,000+
Audit Date	2026-04-26

Overall Risk Score	0.53	MEDIUM

3	8	3	2
Consistent	Mismatches	Over-disclosure	Undeclared

Research prototype · Ongoing research | Policy: https://dassem-ultor-studio.github.io/text-editor-privacy-po...

Executive Summary

Notably Text Editor - Writer & declares 11 data type(s) collected and 10 shared in its Play Data Safety label, across 16 data type(s) analyzed. Our heterogeneous GNN audit model identifies **2 potential undeclared collection(s)** — data types implied by embedded SDK signals but absent from both label and policy — and **8 policy-vs-label mismatch(es)**. Overall risk score: **0.53 (MEDIUM)**.

Top discrepancies by risk:

- **installed_apps**: Undeclared Collection (confidence 0.97)
- **advertising_id**: Undeclared Collection (confidence 0.91)
- **crash_logs**: Policy Label Mismatch (confidence 0.99)

Class definitions:

CONSISTENT: Label, policy, and SDK signals agree.

POLICY_LABEL_MISMATCH: Label discloses data not mentioned in policy.

OVER_DISCLOSURE: Policy mentions data not declared in label.

UNDECLARED_COLLECTION: SDK signals collection undisclosed in label and policy.

Discrepancy Table

Sorted by risk class (undeclared collection first). Y/N: label=data-safety label, policy=policy text, sdk=SDK signal.

Data Type	Class	Conf.	Evidence
installed_apps	**Undeclared Collection**	0.97	label:N policy:N sdk:Y (1, 2, 4 +1)
advertising_id	**Undeclared Collection**	0.91	label:N policy:N sdk:Y (1, 2, 4 +1)
crash_logs	**Policy Label Mismatch**	0.99	label:Y policy:N sdk:Y (1, 2, 4 +1)
other_app_performance_data	**Policy Label Mismatch**	0.99	label:Y policy:N sdk:N (1, 2, 4 +1)
diagnostics	**Policy Label Mismatch**	0.99	label:Y policy:N sdk:Y (1, 2, 4 +1)
in_app_search_history	**Policy Label Mismatch**	0.99	label:Y policy:N sdk:N (1, 2, 4 +1)
approximate_location	**Policy Label Mismatch**	0.98	label:Y policy:N sdk:Y (1, 2, 4 +1)
purchase_history	**Policy Label Mismatch**	0.98	label:Y policy:N sdk:N (1, 2, 4 +1)
user_ids	**Policy Label Mismatch**	0.98	label:Y policy:N sdk:N (1, 2, 4 +1)
precise_location	**Policy Label Mismatch**	0.97	label:Y policy:N sdk:N (1, 2, 4 +1)
date_of_birth	**Over Disclosure**	0.98	label:N policy:Y sdk:N (1, 2, 4 +1)
phone_number	**Over Disclosure**	0.97	label:N policy:Y sdk:N (1, 2, 4 +1)
files_and_docs	**Over Disclosure**	0.96	label:N policy:Y sdk:N (1, 2, 4 +1)
device_id	**Consistent**	1.00	label:Y policy:Y sdk:Y (1, 2, 4 +1)
app_interactions	**Consistent**	1.00	label:Y policy:Y sdk:N (1, 2, 4 +1)
other_info	**Consistent**	1.00	label:Y policy:Y sdk:N (1, 2, 4 +1)

Evidence Trails

Detailed evidence for UNDECLARED_COLLECTION or confidence > 0.85 non-CONSISTENT findings.

Discrepancy: installed_apps — Undeclared Collection (confidence 0.97)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 4, 35

Recommended action: Add 'installed_apps' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

Discrepancy: advertising_id — Undeclared Collection (confidence 0.91)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 4, 35

Recommended action: Add 'advertising_id' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

Discrepancy: crash_logs — Policy Label Mismatch (confidence 0.99)

Label declares collection: **Yes**

Label declares sharing: **Yes**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 4, 35

Recommended action: Update the privacy policy to disclose 'crash_logs' collection, aligning it with the Play Data Safety label.

Discrepancy: other_app_performance_data — Policy Label Mismatch (confidence 0.99)

Label declares collection: **Yes**

Label declares sharing: **Yes**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Update the privacy policy to disclose 'other_app_performance_data' collection, aligning it with the Play Data Safety label.

Discrepancy: diagnostics — Policy Label Mismatch (confidence 0.99)

Label declares collection: **Yes**

Label declares sharing: **Yes**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 4, 35

Recommended action: Update the privacy policy to disclose 'diagnostics' collection, aligning it with the Play Data Safety label.

Discrepancy: in_app_search_history — Policy Label Mismatch (confidence 0.99)

Label declares collection: **Yes**

Label declares sharing: **Yes**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Update the privacy policy to disclose 'in_app_search_history' collection, aligning it with the Play Data Safety label.

Discrepancy: approximate_location — Policy Label Mismatch (confidence 0.98)

Label declares collection: **Yes**

Label declares sharing: **Yes**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 4, 35

Recommended action: Update the privacy policy to disclose 'approximate_location' collection, aligning it with the Play Data Safety label.

Discrepancy: purchase_history — Policy Label Mismatch (confidence 0.98)

Label declares collection: **Yes**

Label declares sharing: **Yes**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Update the privacy policy to disclose 'purchase_history' collection, aligning it with the Play Data Safety label.

Discrepancy: user_ids — Policy Label Mismatch (confidence 0.98)

Label declares collection: **Yes**

Label declares sharing: **Yes**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Update the privacy policy to disclose 'user_ids' collection, aligning it with the Play Data Safety label.

Discrepancy: precise_location — Policy Label Mismatch (confidence 0.97)

Label declares collection: **Yes**

Label declares sharing: **Yes**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Update the privacy policy to disclose 'precise_location' collection, aligning it with the Play Data Safety label.

Discrepancy: date_of_birth — Over Disclosure (confidence 0.98)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Review whether 'date_of_birth' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: phone_number — Over Disclosure (confidence 0.97)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Review whether 'phone_number' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: files_and_docs — Over Disclosure (confidence 0.96)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Review whether 'files_and_docs' is genuinely collected. If yes, add to label; if no, remove from policy.

Methodology

PolicyGraphAudit-RT represents each Android app as a tri-partite heterogeneous graph connecting three layers: **Privacy Policy** text (OPP-115 segment classification), **Play Data Safety Labels** (declared data types and purposes), and **Runtime Evidence** (inferred SDK trackers via Exodus Privacy). A 2-layer HeteroConv R-GCN encodes all three layers jointly; an MLP classifier head predicts discrepancy classes for each (App, DataType) pair.

The model was trained under an edge-masking protocol (30% of label-determining edges removed) to prevent circularity. It achieves **macro F1 = 0.956** on 39 held-out apps (UNDECLARED_COLLECTION F1 = 0.974), trained on 252 apps and 3,202 labeled pairs. Model card: https://github.com/PolicyGraphAudit-RT/reports/blob/main/reports/m5_model_card.md.

Known limitations: (1) Privacy labels are from a 2022 Play Store snapshot. (2) SDK presence is inferred via category-level priors, not measured from APK bytecode. (3) Policy classifier uses MiniLM + logistic regression (OPP-115 macro F1 = 0.70). (4) Labels are rule-derived weak supervision, not human-annotated. (5) English-language Android apps only.

Disclaimer